

PAWAN SINGH KAPKOTI

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PROFESSIONAL SUMMARY

Data & Analytics Engineer with hands-on manufacturing domain experience. Built 4 deployed applications, a GitHub Action used by developers, and contributed to open-source projects with 100K+ combined stars (pandas, ChromaDB, Plotly, Ollama). 36+ automated tests, CI/CD pipelines on every project. MSc Data Analytics (Aston University). Microsoft PL-300 certified.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Databases: PostgreSQL, SQLite, SQL Server, DuckDB

Data Engineering: dbt (53 automated tests), Prefect, GitHub Actions CI/CD, ETL/ELT pipelines

ML / AI: scikit-learn, XGBoost, Ollama (local LLM), ChromaDB (RAG), sentence-transformers

Visualisation: Power BI (PL-300 certified), Streamlit, Plotly, Matplotlib

Cloud: AWS (S3, Glue, Athena, QuickSight), Azure

Tools: Docker, pandas, Git, openpyxl, Flask, React, Firebase

EXPERIENCE

Team Leader | Copernus Fresh Fish, Yorkshire

Apr 2025 – Present

- Led production team of 11+ operatives, coordinating shift patterns and operational data workflows
- Operated SI (Integrator) ERP system for production data management and yield KPI monitoring
- Managed batch traceability reports and weight variance reconciliation for BRC compliance (Lidl, Iceland)

Cover Team Lead | Cranswick Convenience Foods, Yorkshire

Oct 2024 – Apr 2025

- Managed batch-level traceability data for M&S and Tesco premium product lines
- Nav Code and Vector tag validation, allergen cross-reference checks, full audit trail documentation

Food Operative | Cranswick Convenience Foods, Yorkshire

Jul 2024 – Oct 2024

- Collaborated with NPD and Production teams on product specifications and traceability systems

PROJECTS

OpsMind — On-Premises AI Query Tool | Python, Streamlit, Ollama, ChromaDB, SQLAlchemy

- Built NL-to-SQL pipeline enabling non-technical staff to query production databases in plain English, replacing a 2–3 day IT request process
- Designed schema registry mapping 7 business domains to 150 tables, reducing LLM prompt size by 97%
- Implemented password authentication, query caching (5-min TTL), 36 pytest tests, GitHub Actions CI/CD

UK Crime Pipeline — End-to-End Data Pipeline | PostgreSQL, dbt, Streamlit, GitHub Actions

- Architected ELT pipeline: Police UK API ingestion with retry (exponential backoff) into PostgreSQL, dbt transformations, Streamlit dashboard
- Processed 99,675 crime records across 10 UK cities with idempotent upserts; 3 CI/CD workflows (scheduled ingest, tests, health check)

MediAsk — Health Q&A Platform | Flask, PostgreSQL, Gemini AI, Docker

- Built community health platform with NHS-verified guidance, Gemini AI, voice input, 18 languages; Google OAuth; deployed on Render

Apex Data Migration | Prefect, XGBoost, dbt, Power BI

- Simulated on-prem to cloud migration: 10-task Prefect pipeline, XGBoost at 93% accuracy (ROC-AUC 0.97), dbt with 53 tests, Power BI dashboard

UK Education Attainment Analysis | Python, scikit-learn, XGBoost, matplotlib

- ML analysis of 60,896 DfE records across 24 ethnic groups; Gradient Boosting at 94.4% accuracy; 6 publication-ready charts

PSW Visa Tax Analysis | Python, pandas, openpyxl, matplotlib

- Cost-benefit analysis from 20 months of payslip data: net contribution +£29,969; 6-sheet stakeholder Excel report with scenario modelling

Manufacturing Compliance Dashboard | Streamlit, SQLite, Plotly, fpdf2

- Built a BRC/HACCP compliance dashboard for food manufacturing with batch traceability, temperature monitoring, allergen matrix, shelf life concession tracking, and one-click PDF audit export
- SSRS parser auto-detects SI Integreater exports (header rows, merged cells, footers) and maps 70+ column name variations to dashboard schema. Free open-source core + paid premium on Gumroad

SQL Ops Reviewer — GitHub Action | Python, Ollama, GitHub Actions API

- Built an AI-powered GitHub Action that auto-reviews .sql files in pull requests for performance anti-patterns, security risks, and best practices
- Uses Ollama locally — no cloud APIs. Posts structured review comments with findings and fix suggestions directly on the PR

EDUCATION

MSc Data Analytics | Aston University, Birmingham

Jan 2023 – Mar 2024

- Dissertation: Effects of ethnicity on student outcomes — predictive model with 80% accuracy

BCA (Computer Applications) | Amity University, Noida

Aug 2019 – Aug 2022

CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate (PL-300) — March 2026
- Google Data Analytics Professional Certificate
- Azure Data Engineering (DP-203 path)
- AWS Cloud Practitioner

OPEN SOURCE CONTRIBUTIONS

- pandas (45K stars) — PR #65002: Improved str.cat() docstring for Index return type behaviour
- ChromaDB (6K stars) — PR #6819: HNSW tuning guide + PR #6821: Python 3.14 deprecation fix
- ollama-python (5K stars) — PR #638: Fixed ShowResponse ValidationError for cloud models
- Plotly (17K stars) — PR #5559: Added dependabot config for uv.lock dependency updates
- fpdf2 (1K stars) — PR #1803: Fixed TextRegion.In() double line break bug